

Unit - 2

Joint Random variable or Vector.

* Joint Random Variable $\rightarrow$ also Bivariate.

Let 'x' and 'y' be two random variable each assigns a real number in sample space 's', then (x, y) is called joint random variable or two dimensional random variable.

$\Rightarrow$ Let an event 'A' can be defined as random variable 'x' and event 'B' can be defined as random variable 'y'.

$$\text{i.e. } A = \{x \leq x\}$$

$$B = \{y \leq y\}$$

$\Rightarrow$ Random variable $x_1, x_2, \dots, x_n$ is known as n-dimensional random variable.

Cumulative distribution Function (CDF)

$\Rightarrow$ Let (x, y) be a joint random variable, then joint CDF is given as:

$$F_{xy}(x, y) = P[X \leq x, Y \leq y]$$

Properties of CDF:

$$(1) F_{xy}(x, y) \geq 0$$

$$(2) F_{xy}(-\infty, -\infty) = 0$$
$$F_{xy}(\infty, \infty) = 1$$

$$(3) F_{xy}(x, \infty) = F_{xy}(x) / F_x(x)$$

$$F_{xy}(\infty, y) = F_{xy}(y) / F_y(y)$$

$$(4) F_{xy}(-\infty, y) = 0 = F_{xy}(x, \infty)$$

* Joint - Probability density function (Joint - PDF)

⇒ For joint - random variable (x, y) , joint - PDF is given as

$$f_{xy}(x, y) = \frac{d^2}{dx dy} F_{xy}(x, y)$$

Properties of joint - PDF

Area under curve is unity

① $f_{xy}(x, y) \geq 0$

② $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{xy}(x, y) dx dy = 1$

③ $F_{xy}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{xy}(x, y) dx dy$

④ $F_x(x) = \int_{-\infty}^{\infty} \int_{-\infty}^x f_{xy}(x, y) dx dy$

⑤ $F_y(y) = \int_{-\infty}^{\infty} \int_{-\infty}^y f_{xy}(x, y) dx dy$

$\int_{-\infty}^{\infty} f_{xy}(x, y) dx = f_y(y)$

⑥ $P[x_1 < x \leq x_2] = \int_{x_1}^{x_2} f_x(x) dx$

⑦ $P[x_1 < x \leq x_2 : y_1 < y \leq y_2] = \int_{x_1}^{x_2} \int_{y_1}^{y_2} f_{xy}(x, y) dx dy$

⑧ $\int_{-\infty}^{\infty} f_{xy}(x, y) dy = f_x(x)$

(14) $f_{xy}(x, y) = f_x(x) \cdot f_y(y)$ for independent random variables or basically events disjoint

$f_{xyz}(x, y, z) = f_x(x) \cdot f_y(y) \cdot f_z(z)$

(ex1) The Joint PDF of two R.V x, y given by:

$$f_{xy}(x, y) = \begin{cases} C(2x+1) & 0 \leq x \leq 2 \\ & 0 \leq y \leq 3 \\ 0 & \text{o.w} \end{cases}$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{xy}(x, y) dx dy = 1$$

$$\Rightarrow \int_0^2 \int_0^3 C(2x+1) dx dy$$

$$= \int_0^2 C(2x+1) [y]_0^3 dx = 1$$

bas dy
me integrate
and
 $\underbrace{C(2x+1)dx}_{\text{const.}}$

$$= \int_0^2 C(2x+1) 3 dx = 1$$

$$= 3C \int_0^2 2x+1 dx = 1$$

$$= 3C \left[x^2 + x \right]_0^2 = 1$$

$$= 3C (4+2) = 1, \quad \boxed{C = 1/18}$$

(ex2) A Joint-PDF of random variable x, y given as

$$f_{xy}(x, y) = \begin{cases} e^{-(x+y)} & x \geq 0 : y \geq 0 \\ 0 & \text{o.w} \end{cases}$$

(i) $P[x < 1]$

Ans given $f_{xy}(x, y) = e^{-(x+y)} = \underbrace{e^{-x}}_{f(x)} \underbrace{e^{-y}}_{f(y)}$

hence x and y are independent R.V.

Hence for $P[x < 1]$ we can do

$P[x < 1] = -\infty$ to 1 but $x \geq 0$ so

$$P[x < 1] = \int_0^1 e^{-x} dx = -[e^{-x}]_0^1 = [e^{-x}]_0^1$$

(ii) $P[x > y]$

$= 1 - P[x \leq y]$

$$= 1 - \int_0^y \int_{-\infty}^{\infty} f_{xy}(x, y) dx dy = 1 - \int_0^y \int_{-\infty}^{\infty} e^{-x} e^{-y} dx dy$$

Joint moments about origin

Joint moment of random vector $[x, y]$ about origin is given by:

$$m_{nk} = E[x^n y^k] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^n \cdot y^k f_{xy}(x, y) dx dy$$

↳ Joint moments

$\Rightarrow n+k \rightarrow$ order of moment.

$\Rightarrow$ let $k=0$, then $m_{n0} = E[x^n y^0] = E[x^n]$ n^{th} moment of R.V 'x'

$\Rightarrow$ if $n=0$, then

$m_{0k} = E[x^0 y^k] = E[y^k]$, k^{th} moments of R.V 'y'

$\Rightarrow$ if $n=k=1$, then $m_{nk} = m_{11} = E[xy] = \mu_{11} = R_{xy}$

i.e. $R_{xy} = E[xy]$

$\hookrightarrow$ correlation of 'x' and 'y'

$\Rightarrow$ if x and y are ~~correlated~~ uncorrelated or independent

$R_{xy} = E[x] \cdot E[y]$

if $R_{xy} = 0$ -- x & y are orthogonal R.V

Central Joint Moments:

$\rightarrow$ mean

$$m_{nk} = E[(x-\bar{x})^n (y-\bar{y})^k] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x-\bar{x})^n (y-\bar{y})^k f_{xy}(x,y) dx dy$$

$\rightarrow$ mean k about $\bar{x}$ and $\bar{y}$

Case-I : if $n=2, k=0$

$$\mu_{20} = E[(x-\bar{x})^2 (y-\bar{y})^0] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x-\bar{x})^2 (y-\bar{y})^0 f_{xy}(x,y) dx dy$$

$$= E[(x-\bar{x})^2] = \sigma_x^2$$

Case-II if $n=0, k=2$

$$= E[(y-\bar{y})^2] = \sigma_y^2$$

Case - III if $n=k=1$

$$\mu_{11} = E[(x - \bar{x})(y - \bar{y})] = C_{xy} \rightarrow \text{Covariance.}$$

* Relationship b/w R_{xy} & C_{xy}

We know that

$$C_{xy} = E[(x - \bar{x})(y - \bar{y})]$$

$$= E[xy - x\bar{y} - \bar{x}y + \bar{x}\bar{y}]$$

$$E[1] = \int_{-\infty}^{\infty} 1 \cdot f(x) dx$$

$$= \int xy f_{xy} R_{xy} - \bar{y}\bar{x} - \bar{y}\bar{x} + \bar{x}\bar{y}$$

$$E[x]E[y], E[1]$$

$$C_{xy} = R_{xy} - \bar{x}\bar{y}$$

$$C_{xy} = R_{xy} - E[x]E[y]$$

if x and y are independent

$$R_{xy} = E[x] \cdot E[y]$$

$$C_{xy} = 0$$

orthogonal vectors.

Note: Independent R.V, uncorrelated R.V

$x, y \rightarrow$ independent so if it then uncorrelated
 $x, y \rightarrow$ uncorrelated h to zaruri hi independent ho

$$C_{xy} = R_{xy} - E[x] \cdot E[y]$$

if x and y are independent R.V, uncorrelated

$$R_{xy} = E[xy] = E[x] \cdot E[y]$$

$$C_{xy} = E[x] \cdot E[y] - E[x] \cdot E[y] = 0$$

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Co-variance Coefficients (ρ)

$$\rho_{xy} = \rho = \frac{C_{xy}}{\sigma_x^2 \cdot \sigma_y^2}$$

Q1. Let 'x' and 'y' be joint continuous R.V with joint P.d.f is given below.

$$f_{xy}(x, y) = \begin{cases} 2 & x, y > 0 \text{ and } x+y \leq 1 \\ 0 & \text{o.w} \end{cases}$$

- (a) Find - co-variance of 'x' and 'y' (C_{xy})
 (b) " " " coefficient of 'x' & 'y' (ρ)

Ans $C_{xy} = \mu_{xy} - E[X] \cdot E[Y]$

$$\rho = \frac{C_{xy}}{\sigma_x \cdot \sigma_y}$$

$$\Rightarrow C_{xy} = E[XY] - E[X] \cdot E[Y]$$

$$\sigma_x^2 = ?$$

$$\sigma_y^2 = ?$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{xy}(xy) dx dy \quad \left\{ f(xy) \right.$$

$$f_x(x) = \int_{-\infty}^{\infty} f_{xy}(x, y) dy$$

$$f_x(x) = \int_{-\infty}^{\infty} f_{xy}(x, y) dy$$

$$= \int_0^{1-x} 2 dy = 2[1-x]$$

$$f_y(y) = \int_{-\infty}^{\infty} f_{xy}(x, y) dx = \int_0^{1-y} 2 dx = 2[1-y]$$

$$(i) E(x) = \int x f(n) dn = x \int_{-\infty}^{\infty} n 2(1-n) dn$$

$= 2 \int_0^1 (n - \frac{n^2}{2}) dn$

 $= 2 \left[\frac{n^2}{2} - \frac{n^3}{6} \right]_0^1$

 $= 2 \left[\frac{1}{2} - \frac{1}{6} \right]$

 $= 2 \left[\frac{3-1}{6} \right]$

 $= \frac{2 \times 2}{6} = \frac{2}{3}$

$n+y \leq 1$
 $n \text{ max}$
 $y \text{ a sketch}$

$$E(y) = \text{similarly} = \frac{1}{3}$$

$$\text{Ans} = \frac{1}{12}$$

$$(i) E(xy) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{xy}(n, y) dn dy$$

$$= \int_0^1 \int_0^{1-n} 2xy dn dy$$

$$= \int_0^1 2n dn \left[\frac{y^2}{2} \right]_0^{1-n}$$

$$= \int_0^1 2n \frac{(1-n)^2}{2} dn$$

$$= \frac{1}{2} \int_0^1 (n - n^3 - 2n^4) dn$$

$$= \frac{1}{2} \left[\frac{n^2}{2} - \frac{n^4}{4} - \frac{2n^5}{5} \right]_0^1$$

$$\left[\frac{x^2}{2} - \frac{x^4}{4} - \frac{2x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{4} - \frac{2}{3}$$

$$= \frac{6 - 3 - 4}{12} = -\frac{1}{12}$$

$$C_{xy} = R_{xy} - E[X]E[Y]$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 \cdot 2(1-x) dx = \int_0^1 2(x^2 - x^3) dx$$

$$= 2 \left(\frac{x^3}{3} - \frac{x^4}{4} \right)_0^1$$

$$= 2 \left[\frac{1}{3} - \frac{1}{4} \right] = \frac{1}{12} \times 2 = \frac{1}{6} = E[X^2]$$

$$E[Y^2] = \frac{1}{6}$$

$$\sigma_x^2 = \frac{1}{6} - \left(\frac{1}{3} \right)^2 = \frac{1}{18}$$

$$\sigma_y^2 = \frac{1}{18}$$

$$\frac{-1/36}{\sqrt{1/18} \sqrt{1/18}}$$

$$= \frac{-1/36}{1/18} = -\frac{1}{2}$$

$$\rho_{xy} = \frac{C_{xy}}{\sigma_x \sigma_y}$$

$$\sqrt{\sigma_x^2} \sqrt{\sigma_y^2}$$

$$= \frac{-1/36}{1/18} = -\frac{1}{2}$$

Joint Characteristic Function:-

⇒ Joint characteristic function of random variable 'x' and 'y' is given by:-

$$\Phi_n(\omega) = E[e^{j\omega n}]$$

$$\Phi_{xy}(\omega_1, \omega_2) = E[e^{j\omega_1 x} + e^{j\omega_2 y}]$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{j(\omega_1 x + \omega_2 y)} f_{xy}(n, y) dndy$$

⇒ Joint moments of 'x' & 'y' can be written as.

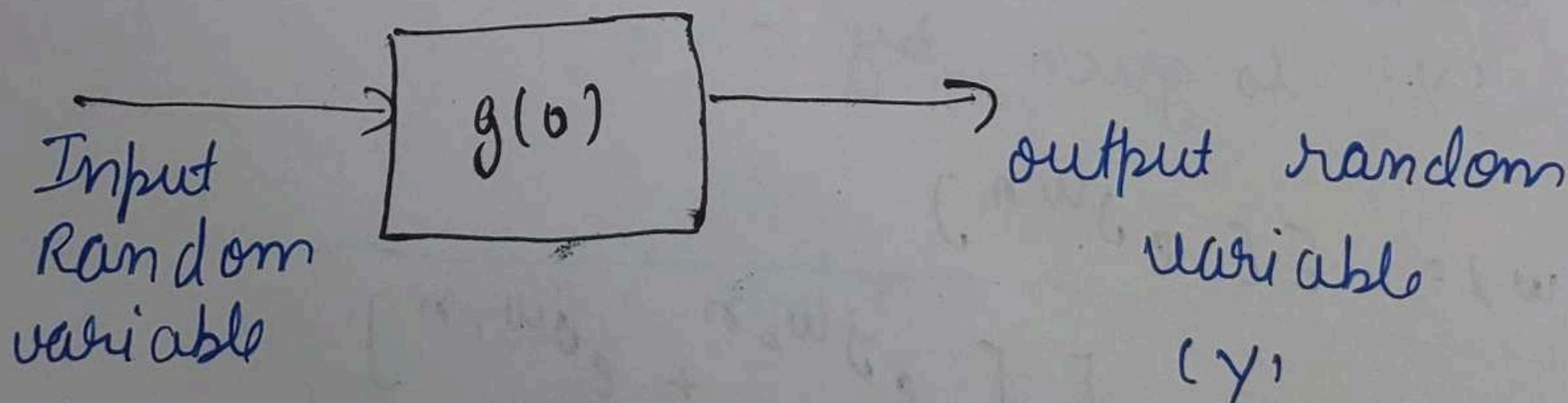
$$m_{nk} = (-j)^{n+k} \frac{\delta^{n+k} (\phi_{xy}(\omega_1, \omega_2))}{d\omega_1^n d\omega_2^k} \quad \left. \begin{array}{l} \text{derivative} \\ \text{w.r.t } \omega \end{array} \right\}$$

ex:- Given that

$$\phi_{xy}(\omega_1, \omega_2) = \text{Expectation } [-2\omega_1^2, -8\omega_2^4]$$

$$m_{11} = ? , E[x] = ? , E[y] = ?$$

Transformation of random variable :



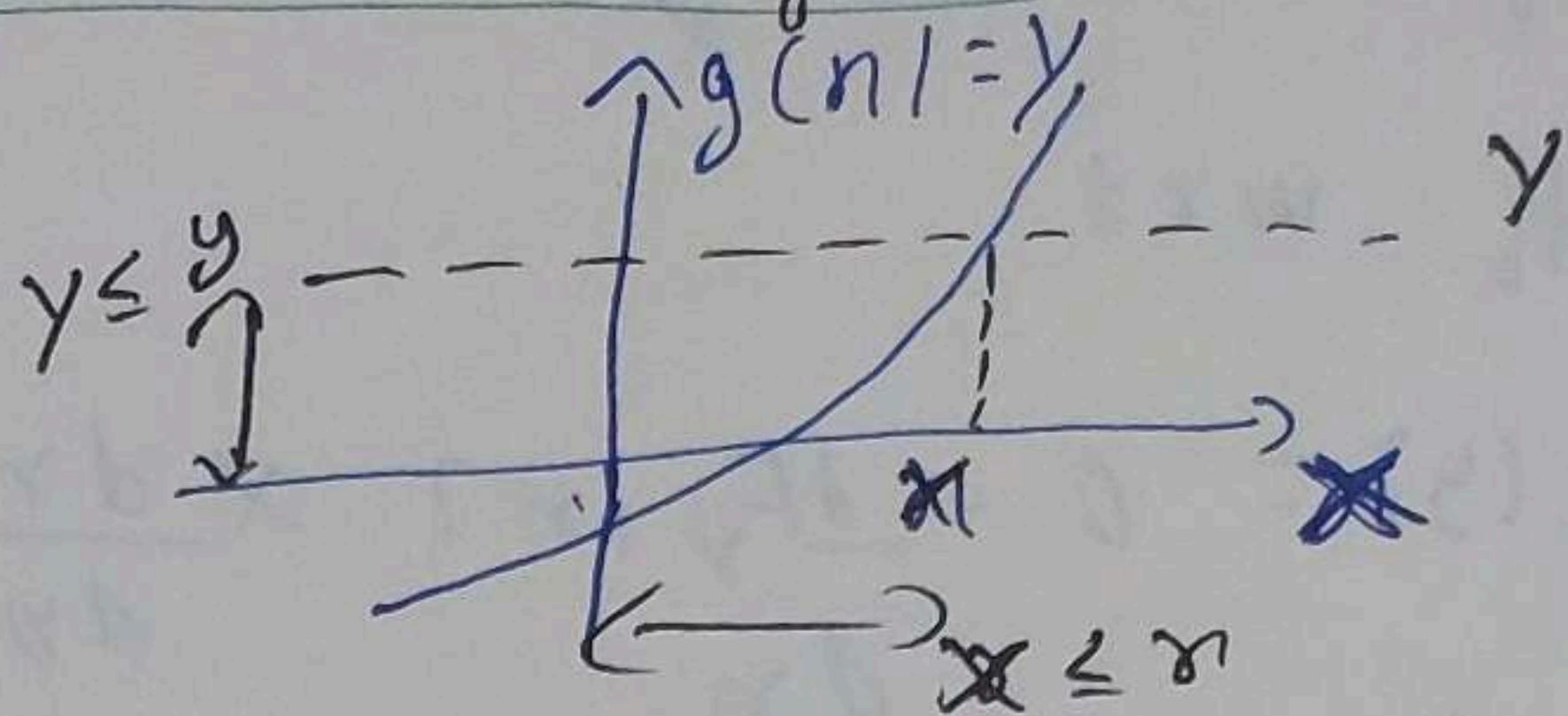
→ CDF $[F_x(n)] \rightarrow$ given
 → PDF $[f_x(n)] \rightarrow$ given

→ CDF $[F_x(x) = ?]$
 → PDF $[f_x(n) = ?]$

Case-I: For monotonically increasing function

$$Y = g(X)$$

$$X = g^{-1}(Y)$$



$$F_Y(y) = P[Y \leq y]$$

$$F_Y(y) = P[X \leq x] = F_X(x)$$

$$F_Y(y) = F_X(x) \quad \text{--- (1)}$$

monotonic k line CDF of x same as CDF of y.

x-ke // line draw, graph ko cut waha se 1x on x-axis

$$\Rightarrow f_Y(y) = \frac{d}{dy} F_Y(y)$$

on differentiating (1) w.r.t 'y'

$$\frac{dF_Y(y)}{dy} = \frac{dF_X(x)}{dx} \cdot \frac{dx}{dy}$$

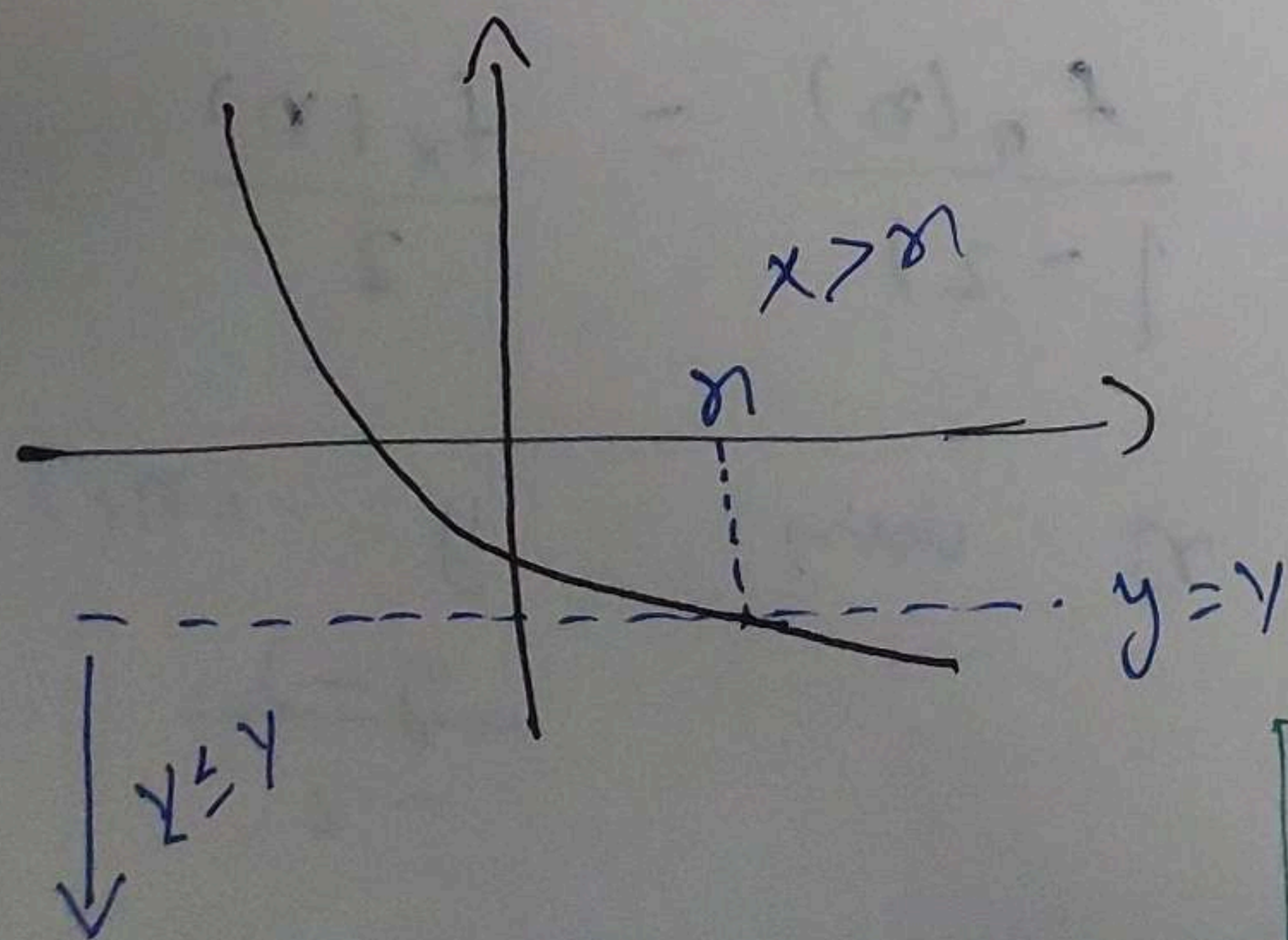
$$\Rightarrow f_Y(y) = f_X(x) \cdot \frac{dx}{dy}$$

relation of CDF.

Relation of PDF

$$f_Y(y) = \frac{f_X(x)}{\frac{dy}{dx}} = \frac{f_X(x)}{\text{slope}}$$

Case-II For Monotonically Decreasing function



same procedure of 1st kheschna

$$F_Y(y) = P[Y \leq y]$$

$$= P[X > x]$$

$$F_Y(y) = 1 - P[X \leq x]$$

equality ni lagati > me

$$F_X(x) + F_Y(y) = 1$$

(CDF)

$$F_Y(y) = 1 - F_X(x)$$

diff wrt

$$f_Y(y) = 0 - \frac{dF_X(x)}{dx} \times \frac{dx}{dy}$$

Relation of CDF

$$f_Y(y) = -f_X(x) \times \text{slope}$$

(ex) let 'x' be a random variable having Gaussian pdf with $\mu=2$, $\sigma^2=4$ and 'y' is another R.V defined as: $y = -2x+3$, find Pdf of R.V y.

Ans.

$$\text{Pdf of } y = \frac{\text{Pdf of } x}{|\text{slope}|}$$

general for
↑ or ↓ turn
slope ka sign end
me kr dena

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}}$$

we know that for gaussian distribution Pdf is given by

$$e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\Rightarrow \frac{1}{2\sqrt{2}\pi} e^{-\frac{1}{2} \times \left(\frac{x-2}{2}\right)^2} = \frac{1}{2\sqrt{2}\pi} e^{-\left(\frac{x-2}{2}\right)^2}$$

$$f_Y(y) = \frac{f_X(x)}{\left(\frac{dy}{dx}\right)} = \frac{f_X(x)}{1-2} = \frac{f_X(x)}{2}$$



now replace x using

$$y = -2x+3$$

$$= \frac{1}{4\sqrt{2}\pi} e^{-\left(\frac{\frac{y-3}{-2}-2}{2}\right)^2}$$

$$\frac{y-3}{-2}$$